2026 Housing EIS — Development Forecast

Project: 2026 Housing Element Environmental Impact Study (EIS)
Purpose: Parcel-level development forecasting for 2035 and 2050 model years to support the Tahoe Regional Transportation Plan (RTP) and Housing EIS alternatives analysis.

Overview

This folder contains the full forecasting workflow for the Lake Tahoe Basin Housing EIS. The workflow is split into two phases: **shared data engineering** (parcel preparation and base unit pool assembly) and **scenario-specific forecasting** (per-alternative unit allocation, TAZ summaries, and travel demand model inputs). Four housing policy alternatives are evaluated, each with its own unit distribution assumptions and known-projects list.

Scenarios at a Glance

Alternative	Name	Total New Units	Key Differentiator
Alternative 1	No Project (RTP/SCS Baseline)	4,319	Replicates the adopted RTP/SCS forecast; no new bonus unit incentives or JADU program
Alternative 2	Housing Forward	8,428	Maximum housing production; full pool conversion, expanded bonus units, JADUs, missing middle
Alternative 3	Housing Preservation	4,319	Same unit total as No Project; achieves housing need through preservation and conversion, not new incentives

Alternative	Name	Total New Units	Key Differentiator
Alternative 4	No Stormwater Change	~5,150	Intermediate production; partial pool conversion and limited JADU program, no stormwater coverage incentives

Scenario Descriptions

Alternative 1 — No Project (RTP/SCS Baseline)

Policy intent: Represents the continuation of existing land use policy with no new housing incentives. This is the baseline against which all other alternatives are measured. Unit assumptions match the 2022 RTP/SCS forecast, updated only for known development activity between 2022 and 2025.

Unit pool summary:

Pool	Units	Where Distributed
Affordable Bonus Units	211	Within Bonus Unit Boundary
Moderate Bonus Units	271	Within Bonus Unit Boundary
Achievable Bonus Units	20	Within Bonus Unit Boundary
Affordable by Design	0	—
JADUs	0	—
Private CFA/TAU Conversion	700	Anywhere residential allowed
Banked Residential Units	399	Anywhere residential allowed
Local Jurisdiction BU Pool (unassigned)	238	Model assigns
Local Jurisdiction BU Pool (assigned projects)	427	Model assigns
Remaining 2012 RP Allocations	1,956	Model assigns
Total New Units	**4,319**	

Key assumptions:

- No conversion of unused CFA/TAU/PAOT development rights to bonus units
- No Junior ADU (JADU) program
- Affordable/Moderate bonus units directed entirely within the Bonus Unit Boundary to capture fee waivers
- Remaining 2012 Regional Plan allocations: 33% multifamily (100% occupied), 66% single-family
- ADU pool = 110 units (no Washoe County expansion beyond what the updated ADU condition allows)
- TRPA Bonus Unit pool = 388; TRPA General pool = 948

Changes from RTP model run (common to all alternatives):

1. Known residential unit changes from 2022–2025 pre-incorporated into the input parcel pickle
2. Zoned unit pools adjusted for construction activity 2022–2025 (per K. Kasman)
3. Updated known projects list
4. ADUs now allowed in Washoe County (ADU eligibility condition updated)
5. 33% of allocations assumed multifamily (100% occupied); remaining 66% single-family
6. Population in 2035 and 2050 allowed to vary based on persons per household and occupancy rate

Alternative 2 — Housing Forward

Policy intent: Emphasizes housing production and preservation outcomes by strengthening incentives for workforce and deed-restricted housing, expanding the scope of bonus unit and fee reforms, and activating new unit categories (JADUs, missing middle) while maintaining required environmental protections and growth limits. Adds approximately 3,316 units above the No Project baseline (net 8,428 total new units).

Unit pool summary:

Pool	Units	Where Distributed
Affordable Bonus Units	1,527	Within Bonus Unit Boundary
Moderate Bonus Units	955	Within Bonus Unit Boundary
Achievable Bonus Units	764	65% within BUB / 35% anywhere
Affordable by Design	382	Anywhere residential allowed

Pool	Units	Where Distributed
JADUs	792	Anywhere residential allowed
Private CFA/TAU Conversion	700	Anywhere residential allowed
Banked Residential Units	399	Anywhere residential allowed
Local Jurisdiction BU Pool (unassigned)	238	Anywhere residential allowed
Local Jurisdiction BU Pool (assigned projects)	427	Model assigns
Remaining 2012 RP Allocations	1,956	Model assigns
Achievable General Pool	191	Anywhere residential allowed
Total New Units	**8,428**	

Key assumptions:

- Full conversion of unused CFA/TAU/PAOT development rights to residential bonus units, generating substantially larger Affordable (1,527), Moderate (955), and Achievable (955) pools
- Active JADU program: 32 units/year from 2027–2050 = 792 JADUs distributed throughout the region
- Affordable by Design bonus units (382) distributed anywhere residential development is allowed
- 65% of Achievable Bonus units within Bonus Unit Boundary; 35% anywhere
- Remaining jurisdictional pool units distributed 100% anywhere residential development is allowed (not constrained to MF parcels)
- TRPA Bonus Unit pool = 947; TRPA General pool = 1,174; ADU pool = 150
- Missing middle housing allowed in more neighborhoods

Alternative 3 — Housing Preservation

Policy intent: Focuses on achieving housing need through the **preservation and conversion of existing housing** to workforce housing, rather than new production incentives. Requires that a defined percentage of a jurisdiction's housing stock is owned or rented by local workers. Adopts only a subset of proposed policy components — without zoning incentives, bonus unit reforms, coverage incentives, excess coverage mitigation fee changes, or new JADU incentives. No mapping changes from the No Project alternative. Unit total is the same as Alternative 1 (4,319), but with slightly more flexible spatial distribution.

Unit pool summary:

Pool	Units	Where Distributed
Affordable Bonus Units	211	Within Bonus Unit Boundary
Moderate Bonus Units	271	Within Bonus Unit Boundary
Achievable Bonus Units	20	Within Bonus Unit Boundary
Affordable by Design	0	—
JADUs	0	—
Private CFA/TAU Conversion	700	Anywhere residential allowed
Banked Residential Units	399	Anywhere residential allowed
Local Jurisdiction BU Pool (unassigned)	238	Anywhere residential allowed
Local Jurisdiction BU Pool (assigned projects)	427	Model assigns
Remaining 2012 RP Allocations	1,956	Model assigns
Total New Units	**4,319**	

Key assumptions:

- No conversion of unused development rights pools to bonus units
- No new JADU incentives
- No coverage incentives for multifamily or changes to the Excess Coverage Mitigation Fee
- No new incentives for missing middle housing
- Bonus unit and general pool sizes match the updated (post-2025) jurisdiction pools but without new Affordable/Moderate/Achievable categories
- TRPA Bonus Unit pool = 947; TRPA General pool = 1,174; ADU pool = 150
- Achievable Bonus distribution set within Bonus Unit Boundary (same as No Project)
- Spatial distribution of remaining pools is slightly more flexible than Alt 1 (1,337 units "anywhere" vs. 1,099), reflecting WTAP passage allowing Washoe County ADUs
- Sets an explicit target for new occupied residential units (model calibrates to this target)

Alternative 4 — No Stormwater Change

Policy intent: Excludes changes to **coverage limits for multifamily housing** and changes to **excess coverage mitigation fees** — focusing incentives in areas with existing area-wide stormwater treatment. Still provides some support for workforce housing through partial conversion of unused development rights to residential bonus units and expanding uses of bonus units. Some zoning incentives apply (e.g., missing middle in more neighborhoods), but more limited scope than Housing Forward. Up to a fourplex allowed in any residential zone within ½ mile of transit, a town center, or within multifamily zones. Jurisdictions may optionally update area plans for Phase 2 height/density/coverage/parking incentives for Tier 3 housing. Unit total falls between the No Project and Housing Forward alternatives.

Unit pool summary:

Pool	Units	Where Distributed
Affordable Bonus Units	258	Within Bonus Unit Boundary
Moderate Bonus Units	161	Within Bonus Unit Boundary
Achievable Bonus Units	161	65% within BUB / 35% anywhere
Affordable by Design	64	Anywhere residential allowed
JADUs	154	Anywhere residential allowed
Private CFA/TAU Conversion	700	Anywhere residential allowed
Banked Residential Units	399	Anywhere residential allowed
Local Jurisdiction BU Pool (unassigned)	238	Anywhere residential allowed
Local Jurisdiction BU Pool (assigned projects)	427	Model assigns
Remaining 2012 RP Allocations	1,956	Model assigns
Achievable General Pool	32	Anywhere residential allowed
Total New Units	**~5,150**	

Key assumptions:

- Partial conversion of unused CFA/TAU/PAOT development rights — smaller Affordable (258), Moderate (161), and Achievable (161) pools than Housing Forward

- Limited JADU program: 154 JADUs (roughly 6.5 units/year from 2027–2050) distributed throughout the region
- No stormwater coverage incentives for multifamily; no changes to Excess Coverage Mitigation Fee
- Bonus unit and general pool sizes match the updated jurisdiction pools (same as Alt 2/3)
- TRPA Bonus Unit pool = 947; TRPA General pool = 1,174; ADU pool = 150
- Fourfold zoning: up to fourplex allowed near transit/town centers even without coverage incentives
- Tweaks the unit target input CSV relative to No Project to model the partial additional production

Scenario Comparison — Key Differences

Feature	Alt 1 (No Project)	Alt 2 (Housing Forward)	Alt 3 (Preservation)	Alt 4 (No Stormwater)
CFA/TAU/PAOT Pool Conversion	No	Yes (full)	No	Yes (partial)
JADU Program	No	Yes (792 units)	No	Yes (154 units)
Affordable BU Pool	211	1,527	0 new	258
Moderate BU Pool	271	955	0 new	161
Achievable BU Pool	20	764	0 new	161
Affordable by Design	0	382	0	64
Coverage Incentives for MF	N/A	Yes	No	No
Missing Middle Zoning	No	Yes (broad)	No	Yes (limited, near transit)
Preservation/Conversion Requirement	No	No	Yes	No
TRPA Bonus Unit Pool	388	947	947	947

Feature	Alt 1 (No Project)	Alt 2 (Housing Forward)	Alt 3 (Preservation)	Alt 4 (No Stormwater)
TRPA General Pool	948	1,174	1,174	1,174
ADU Pool	110	150	150	150
Total New Units	**4,319**	**8,428**	**4,319**	**~5,150**

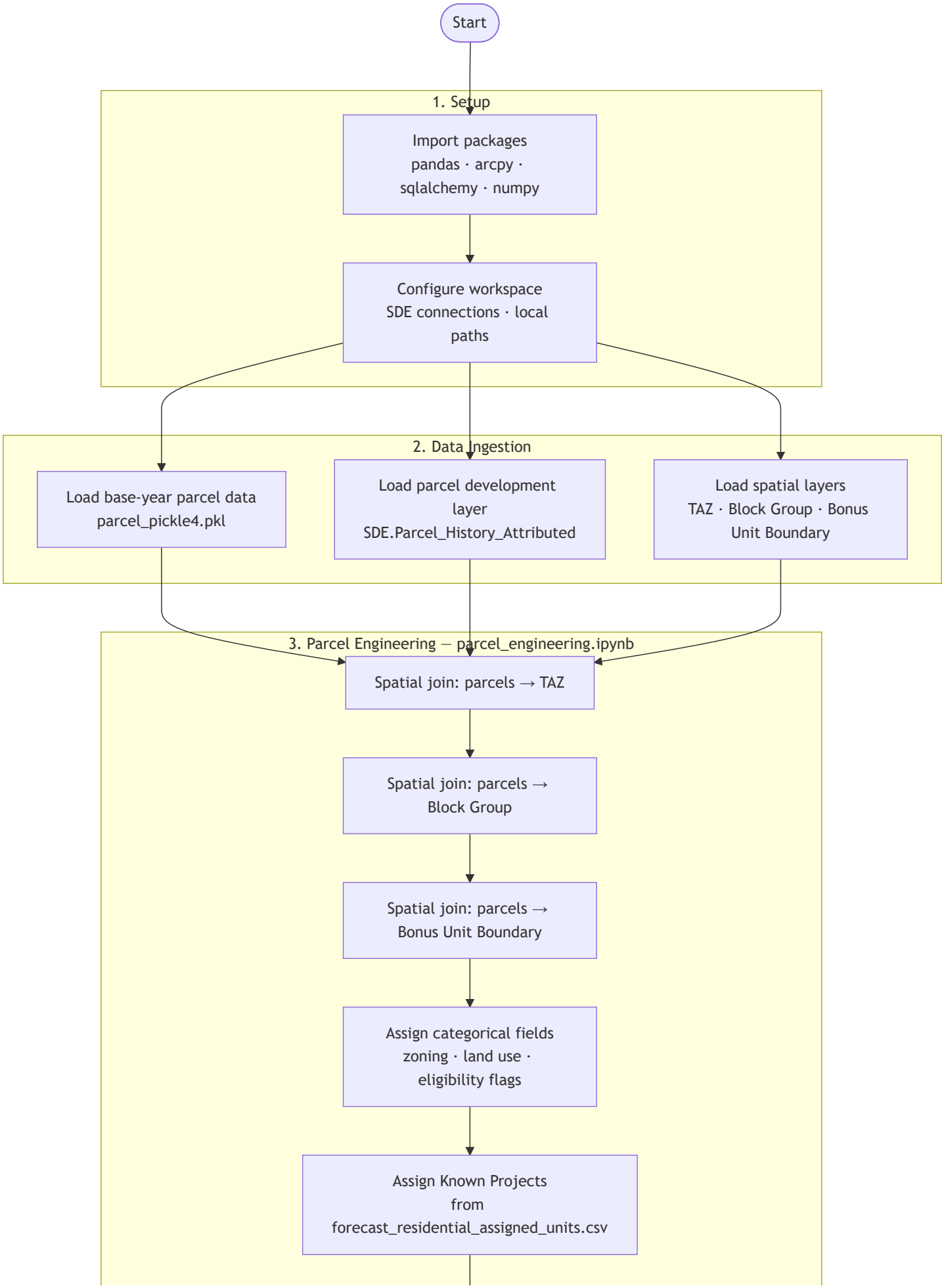
Folder Structure

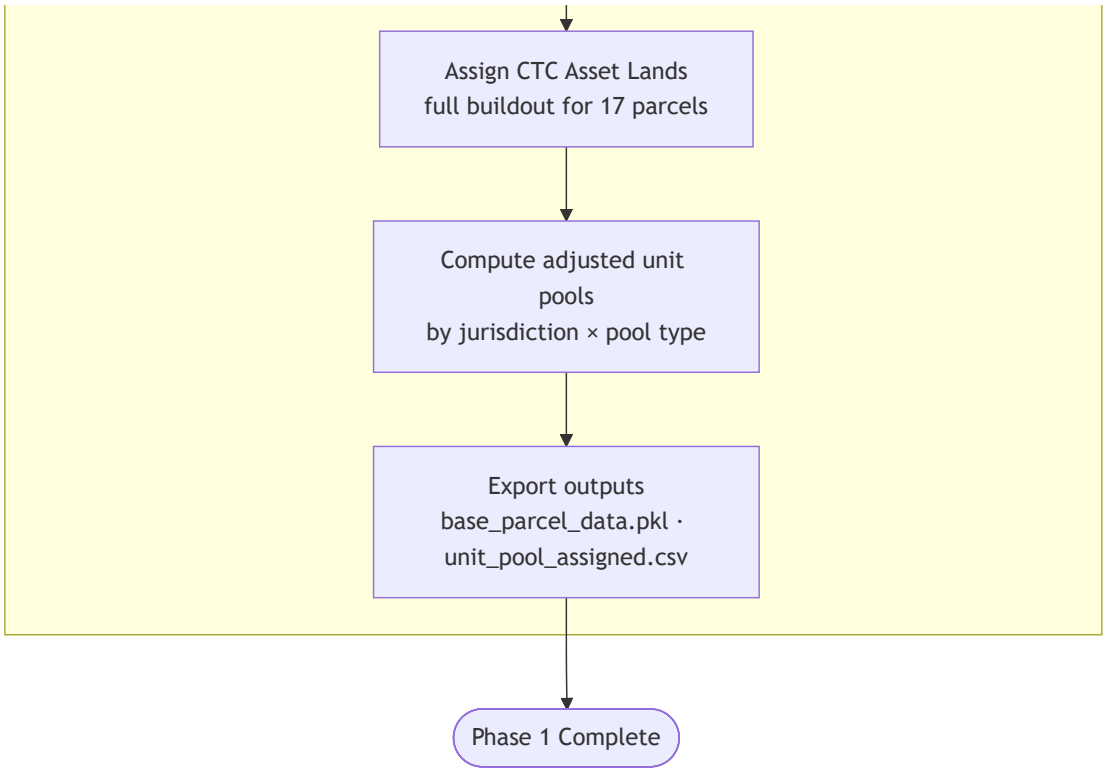
```
Forecast/
├── README.md                          # This file
├── scripts/                           # Shared Phase 1 scripts (run once, common to all alterna
│   ├── parcel_engineering.ipynb       # Phase 1A: spatial joins, parcel classification, unit po
│   ├── scenario_forecast_template.ipynb  # Phase 1B: reference template for scenario notebooks
│   ├── Base_Forecast.ipynb            # Original combined notebook (source of Phase 1A/1B)
│   ├── utils.py                       # Shared helper functions
│   └── Lookup_Lists/                  # Shared input CSVs (zoned units, assigned units, CTC lan
├── data/                              # Shared processed data outputs
│   ├── base_parcel_data.pkl           # Engineered parcel layer (output of Phase 1A)
│   ├── unit_pool_assigned.csv         # Adjusted unit pools after known-project subtraction
│   ├── taz_summary_2035.csv/.pickle   # TAZ-level summary — 2035 model year
│   ├── taz_summary_2050.csv/.pickle   # TAZ-level summary — 2050 model year
│   ├── Parcels_Forecast.csv           # Parcel-level forecast (tabular)
│   └── TAZ_Units.csv                  # Existing and forecasted residential units by TAZ
├── Alternative_1/                     # Housing Alternative 1: No Project (RTP/SCS Baseline)
│   ├── Alternative_1_Forecast.ipynb
│   ├── utils.py
│   ├── inputs/                        # Alternative-specific zoned units and scenario distribut
│   └── Changes from RTP Model Run.md
├── Alternative_2/                     # Housing Alternative 2: Housing Forward
│   ├── Alternative_2_Forecast.ipynb
│   ├── inputs/
│   └── Changes from RTP Model Run.md
├── Alternative_3/                     # Housing Alternative 3: Housing Preservation
│   ├── Alternative_3_Forecast.ipynb
│   ├── inputs/
│   └── Changes from RTP Model Run.md
└── Alternative_4/                     # Housing Alternative 4: No Stormwater Change
    ├── Alternative_4_Forecast.ipynb
    ├── inputs/
    └── Changes from RTP Model Run.md
```

Process Flow

Phase 1 — Shared Data Engineering (`scripts/`)

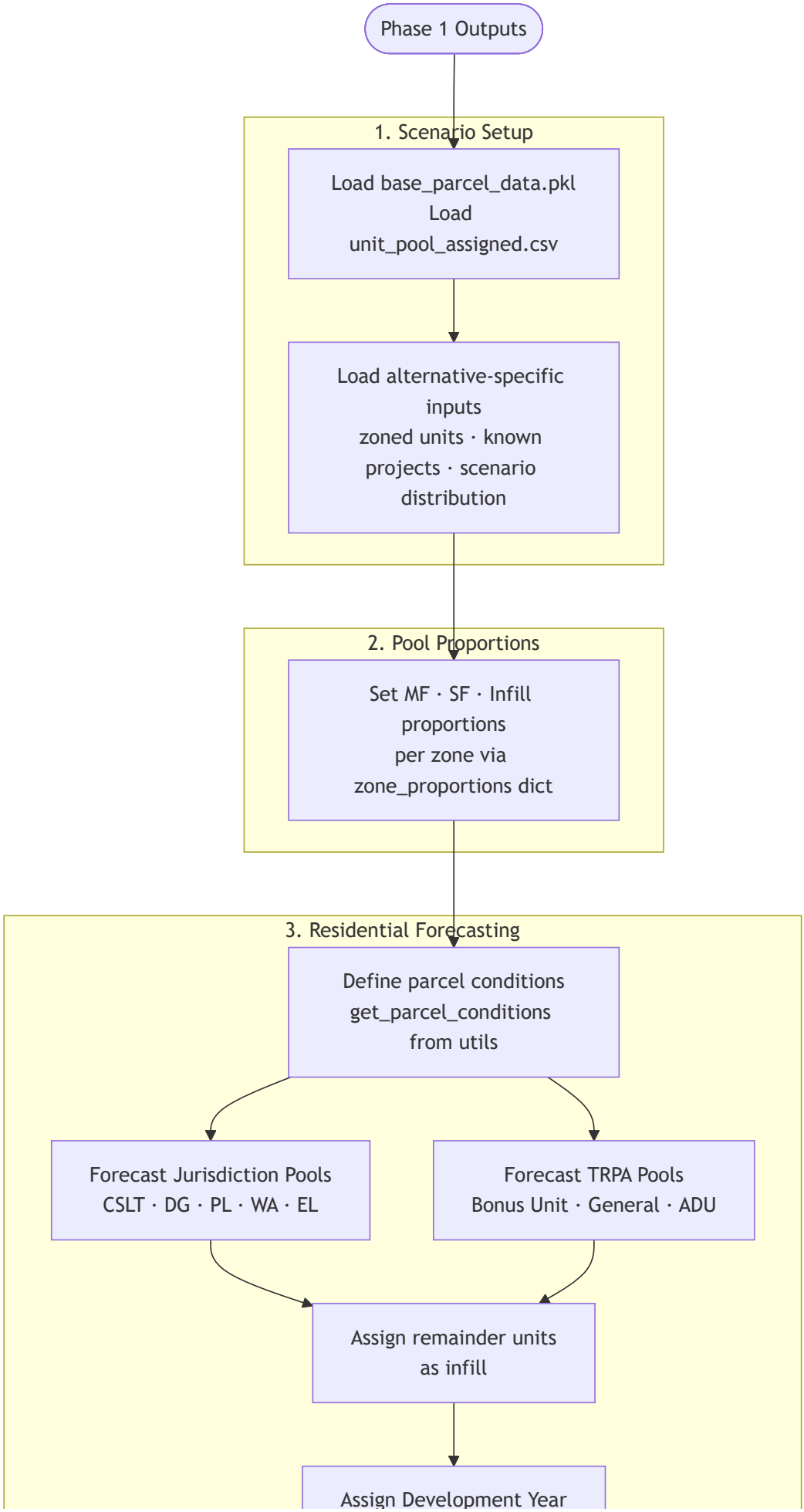
Run once. Outputs are shared across all alternatives.

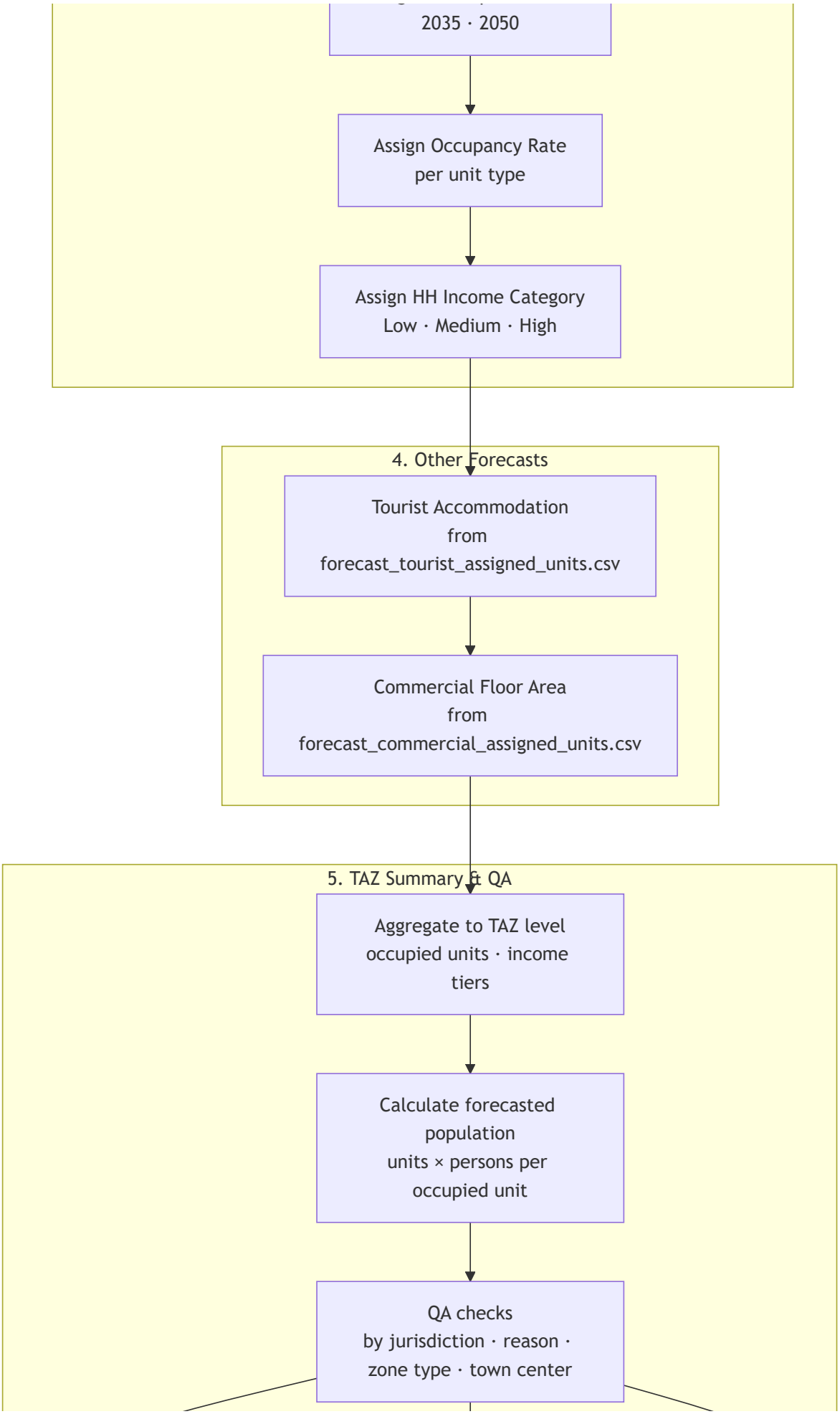

Start
1. Setup
Import packages
pandas · arcpy ·
sqlalchemy · numpy
Configure workspace
SDE connections · local
paths
2. Data Ingestion
Load base-year parcel data
parcel_pickle4.pkl
Load parcel development
layer
SDE.Parcel_History_Attributed
Load spatial layers
TAZ · Block Group · Bonus
Unit Boundary
3. Parcel Engineering — parcel_engineering.ipynb
Spatial join: parcels → TAZ
Spatial join: parcels →
Block Group
Spatial join: parcels →
Bonus Unit Boundary
Assign categorical fields
zoning · land use ·
eligibility flags
Assign Known Projects
from
forecast_residential_assigned_units.csv

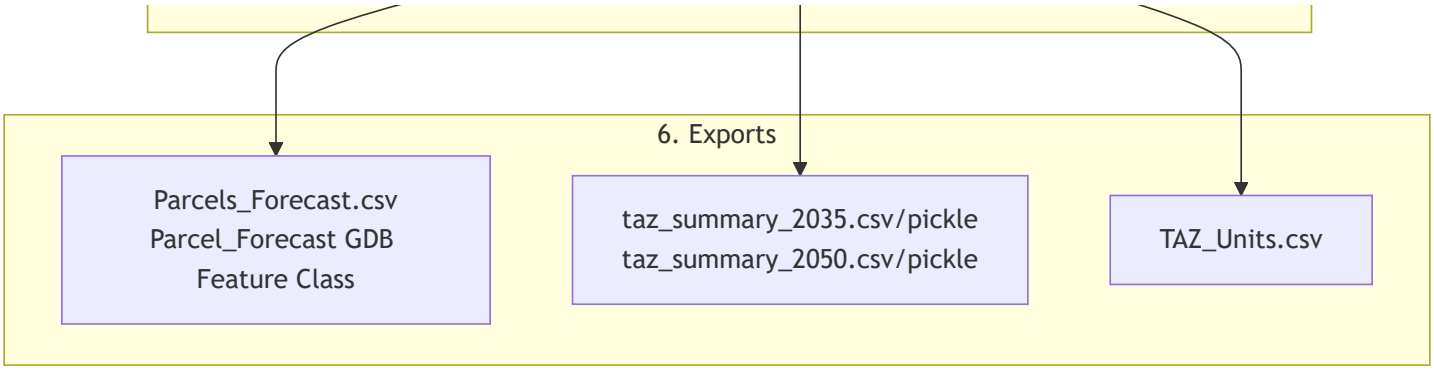


Phase 2 — Scenario Forecasting (per Alternative)

Run independently for each alternative using its own `inputs/` folder.


Phase 1 Outputs
1. Scenario Setup
Load base_parcel_data.pkl
Load
unit_pool_assigned.csv
Load alternative-specific
inputs
zoned units · known
projects · scenario
distribution
2. Pool Proportions
Set MF · SF · Infill
proportions
per zone via
zone_proportions dict
3. Residential Forecasting
Define parcel conditions
get_parcel_conditions
from utils
Forecast Jurisdiction Pools
CSLT · DG · PL · WA · EL
Forecast TRPA Pools
Bonus Unit · General · ADU
Assign remainder units
as infill
Assign Development Year


2035 · 2050
Assign Occupancy Rate per unit type
Assign HH Income Category Low · Medium · High
4. Other Forecasts
Tourist Accommodation from forecast_tourist_assigned_units.csv
Commercial Floor Area from forecast_commercial_assigned_units.csv
5. TAZ Summary & QA
Aggregate to TAZ level occupied units · income tiers
Calculate forecasted population units × persons per occupied unit
QA checks by jurisdiction · reason · zone type · town center



Notebooks

Notebook	Phase	Purpose
`scripts/parcel_engineering.ipynb`	1A	Spatial joins, parcel classification, unit pool assembly; exports `base_parcel_data.pkl` and `unit_pool_assigned.csv`
`scripts/scenario_forecast_template.ipynb`	1B	Reference template for building alternative notebooks; loads Phase 1A outputs
`scripts/Base_Forecast.ipynb`	—	Original combined notebook (source for the Phase 1A/1B split); kept for reference
`Alternative_1/Alternative_1_Forecast.ipynb`	2	Scenario forecast for Alternative 1 — No Project
`Alternative_2/Alternative_2_Forecast.ipynb`	2	Scenario forecast for Alternative 2 — Housing Forward
`Alternative_3/Alternative_3_Forecast.ipynb`	2	Scenario forecast for Alternative 3 — Housing Preservation
`Alternative_4/Alternative_4_Forecast.ipynb`	2	Scenario forecast for Alternative 4 — No Stormwater Change

Inputs

Shared Inputs (`scripts/Lookup_Lists/`)

Input	Type	Description
`forecast_residential_assigned_units.csv`	CSV	Known residential allocations 2023–2025
`forecast_residential_zoned_units.csv`	CSV	Jurisdiction-level zoned residential unit pools
`forecast_tourist_assigned_units.csv`	CSV	Known tourist accommodation changes
`forecast_tourist_zoned_units.csv`	CSV	Jurisdiction-level zoned tourist unit pools
`forecast_commercial_assigned_units.csv`	CSV	Known commercial floor area changes
`forecast_commercial_zoned_units.csv`	CSV	Jurisdiction-level zoned commercial floor area
`CTC_AssetLands_Lookup.csv`	CSV	17 CTC asset land parcels for full buildout
`known_projects.csv`	CSV	Master list of known development projects
`SocioEcon_Summer.csv`	CSV	Base-year TAZ socioeconomic data (persons per unit)

Alternative-Specific Inputs (`Alternative_N/inputs/`)

Input	Type	Description
`forecast_residential_zoned_units.csv`	CSV	Scenario-adjusted zoned unit pool overrides
`known_projects.csv`	CSV	Alternative-specific known projects list
`scenarioN_instructions.csv`	CSV	Narrative description of scenario assumptions
`scenarioN_unit_distribution.csv`	CSV	Per-pool unit counts and spatial distribution targets

Spatial / Database Inputs

Input	Type	Description
`parcel_pickle4.pkl`	Pickle	Base-year parcel data from the 2022 Travel Demand Model
`SDE.Parcel_History_Attributed`	SDE Feature Class	Parcel development records with attributed history
TAZ polygons	SDE Feature Class	Traffic Analysis Zones
Block Group polygons	SDE Feature Class	Census block groups
Bonus Unit Boundary	SDE Feature Class	TRPA bonus unit eligibility boundary

Outputs

Shared (`data/`)

Output	Type	Description
`base_parcel_data.pkl`	Pickle	Engineered parcel layer with spatial joins and classification (Phase 1A output)
`unit_pool_assigned.csv`	CSV	Unit pools after subtracting known projects and CTC lands
`Parcels_Forecast.csv`	CSV	Parcel-level forecast with all assigned attributes
`taz_summary_2035.pkl/.csv`	Pickle / CSV	TAZ-level summary for 2035 model year
`taz_summary_2050.pkl/.csv`	Pickle / CSV	TAZ-level summary for 2050 model year
`TAZ_Units.csv`	CSV	Existing and forecasted residential units by TAZ

Per Alternative

Output	Type	Description
`Parcel_Forecast`	GDB Feature Class	Spatial parcel forecast for GIS use
`taz_summary_2035.csv/.pickle`	CSV / Pickle	Alternative-specific TAZ summary — 2035
`taz_summary_2050.csv/.pickle`	CSV / Pickle	Alternative-specific TAZ summary — 2050

Key Logic

Two-Phase Architecture

The workflow is intentionally split so that the computationally expensive parcel engineering (spatial joins, SDE reads) only runs once. All four alternatives share the same engineered parcel layer and start from the same adjusted unit pools.

Phase	Script	Runs	Key Output
1A — Parcel Engineering	`parcel_engineering.ipynb`	Once	`base_parcel_data.pkl` , `unit_pool_assigned.csv`
1B — Scenario Template	`scenario_forecast_template.ipynb`	Reference only	—
2 — Scenario Forecast	`Alternative_N_Forecast.ipynb`	Once per alternative	TAZ summaries, parcel forecast

Residential Unit Allocation

Units are allocated in a strict priority order:

1. **Known Projects** — directly mapped by APN from the assigned units lookup

2. **CTC Asset Lands** — full buildout assigned to 17 specific parcels
3. **Jurisdictional Pools** — remaining units distributed by jurisdiction (CSLT, DG, PL, WA, EL) and unit type via the `zone_proportions` dictionary:
   - Multifamily (MF) — default 35% of pool (overridable per zone)
   - Single-family (SF) — default 50% of pool (overridable per zone)
   - Infill — default 15% of pool (overridable per zone)
4. **TRPA Pools** — bonus unit and general allocations administered by TRPA
5. **Remainders** — leftover pool units assigned as infill
6. **ADUs** — Accessory Dwelling Units filled to reach regional total

Pool Proportion Overrides

Each alternative notebook contains a `zone_proportions` dictionary keyed by `(Jurisdiction, Unit_Pool)`. Any zone not listed falls back to global defaults. This allows fine-grained scenario differentiation without changing core allocation logic.

```
zone_proportions = {
    ('CSLT', 'Bonus Unit'): {'mf': 0.50, 'sf': 0.35, 'infill': 0.15},
    # ... more overrides ...
    'default': {'mf': 0.35, 'sf': 0.50, 'infill': 0.15},
}
```

Development Year Assignment

Year	Share	Method
2035	~33% of total	All "Assigned" projects + proportional random draw
2050	Remainder	Remaining parcels

Parcel Eligibility Conditions

Parcels are filtered by jurisdiction-specific criteria before unit assignment (defined in `utils.get_parcel_conditions()`):

- **Vacant buildable:** private, not retired, IPES score > 0 (Placer: > 726)
- **Infill:** existing residential with specific zoning
- **Bonus unit boundary:** must be within the TRPA-designated polygon
- **ADU eligible:** existing single residential unit with ADU allowed flag set

Population Calibration

Forecasted population = occupied units × persons per occupied unit (from base-year socioeconomic data). The persons-per-unit factor is adjusted iteratively until regional population targets are met:

- **2035 target:** 55,592 persons
- **2050 target:** 57,611 persons

Key Changes from RTP Model Run

The following changes are common across all alternatives (documented in each `Changes from RTP Model Run.md`):

1. Known residential unit changes from 2022–2025 are pre-incorporated into the input parcel pickle
2. Adjusted zoned unit pools to account for construction between 2022 and 2025 (per K. Kasman)
3. Updated known projects list
4. ADUs are now allowed in Washoe County
5. 33% of allocations assumed to be multi-family (100% occupied); remaining 66% single-family
6. Population in 2035 and 2050 is allowed to vary based on persons per household and occupancy rate (not fixed to a target)

Dependencies

Package	Use
`pandas`	Data manipulation
`arcpy`	Spatial joins, SDE reads, feature class export
`numpy`	Numeric operations
`sqlalchemy`	SQL Server database connections
`pathlib`	File path management
`pickle`	Intermediate data serialization

Package	Use
`utils.py`	Project-specific helper functions (`get_parcel_conditions` , `forecast_residential_units` , `forecast_residential_units_infill` , `get_target_sum` , `check_parcel_condition` , etc.)

> **Environment:** Requires an ArcGIS Pro Python environment with access to the TRPA SDE geodatabase and SQL Server instances (`sql12` , `sql14`). Database credentials are read from the `DB_USER` and `DB_PASSWORD` environment variables.